

CS313: Information Retrieval

Week 12: Future of Search

Multimodal, Agentic, & Neuro-Symbolic AI



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Course Recap: The Big Picture



Classic IR

Weeks 1-5

Boolean, Vector Space, TF-IDF, Inverted Indices.



Web Scale

Weeks 6-8

PageRank, Crawling, Distributed Indexing, Evaluation.



Neural & AI

Weeks 9-12

Embeddings, RAG, Agents, Multimodal Search.

*"We moved from **matching keywords** to **understanding intent**."*

Agenda



Multimodal Search

CLIP, Vision Transformers, and searching beyond text.



Agentic Search

From retrieving information to completing tasks (ReAct).



Neuro-Symbolic AI

Combining LLMs with Knowledge Graphs (GraphRAG).



Course Recap

Connecting the dots: TF-IDF $\rightarrow$ PageRank $\rightarrow$ Neural IR.

Multimodal Search



Beyond Text: The **Multimodal** Shift



The Old Way

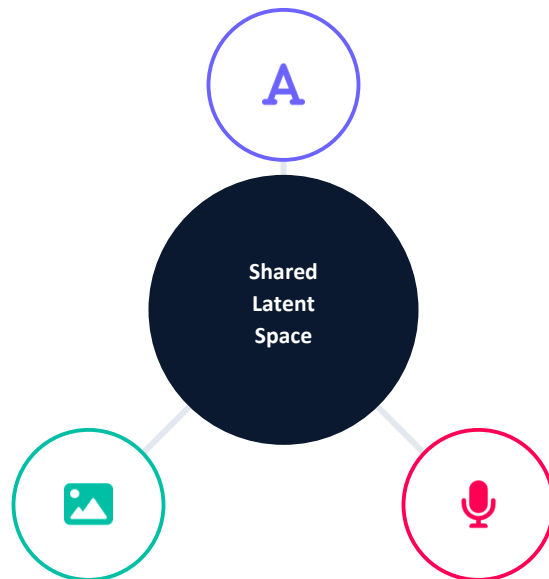
"10 Blue Links". Text in, Text out. Relying on metadata tags for images and video.

The New Way

Searching **pixels** and **audio waves** directly. Understanding content without text descriptions.

Unified Embedding

Mapping images, text, and audio into the **same vector space**.



Contrastive Language–Image Pre-training CLIP

Contrastive Pre-training



Training Objective

Maximize the cosine similarity of correct (image, text) pairs (diagonal) while minimizing incorrect ones.

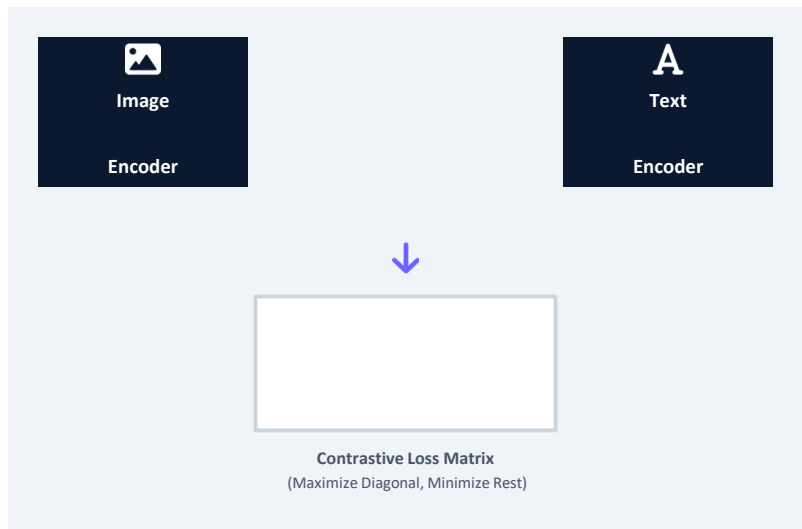
Zero-Shot Ability

Can classify images into categories it has never seen before by prompting: "A photo of a [CLASS]".

Data Scale

Trained on 400 million millions of image-caption pairs from the internet (WebImageText).

A Famous AI model from [OpenAI](#) that learns the relationship between images and text together



Contrastive Language–Image Pre-training CLIP

Contrastive Pre-training



Why CLIP Became Important?

CLIP enabled many modern AI systems:

- image search
- text-to-image generation
- multimodal AI
- vision-language systems
- image understanding without task-specific training

Conclusion

CLIP learns to connect images and language by pulling matching image-text pairs closer together in embedding space and pushing mismatched pairs apart.

A Famous AI model from [OpenAI](#) that learns the relationship between images and text together



ViT: Vision Transformers



✂ Patches as Tokens

Instead of words, we split an image into fixed-size patches (e.g., 16x16 pixels).

≡ Flatten & Project

Each patch is flattened into a vector, similar to a word embedding in NLP.

⚡ Self-Attention

The Transformer learns relationships between distant parts of the image (Global Context).

Example:

the model can connect a face with a body or wheels with a car

Input Image

1	2	3
4	5	6
7	8	9



Sequence of Patches

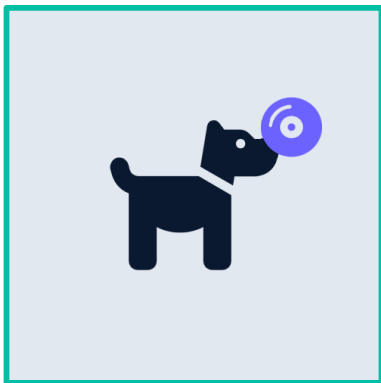


Transformer Encoder

Demo: Zero-Shot Retrieval



 a dog playing frisbee



Score: 0.89

No metadata tags used.



Score: 0.12

Semantic mismatch.



Score: 0.05

Completely unrelated.

Video Retrieval: Frame-Level Search

🕒 Temporal Localization

Users don't just want the video; they want the exact moment (e.g., "04:12").

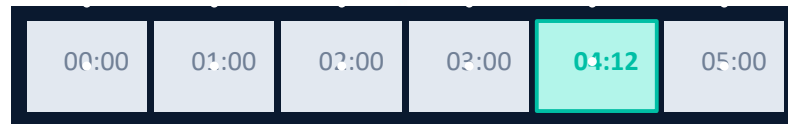
🖼️ Frame Extraction

Videos are treated as a sequence of images. We sample frames (e.g., 1 fps) and embed them using CLIP.

🔍 The Search

Query embedding is compared against **all frame embeddings** to find the highest similarity score.

🔍 "Find the car explosion"



Match Found: 04:12

Audio Search & Whisper



⌵ The Challenge

Audio is opaque to traditional search engines. We need to convert it into a searchable format.

🗄️ OpenAI Whisper

A robust ASR (Automatic Speech Recognition) model trained on 680,000 hours of multilingual data.

🔍 Retrieval

Once transcribed, audio becomes text. We can then apply standard IR techniques (BM25, Embeddings).



Raw Audio

Podcast, Meeting, Video



Whisper Model

Speech-to-Text



Transcript

Searchable Text Index

Agentic Search



From Search to Action



Information Retrieval

Goal	Find Documents
Interaction	Read-Only
Output	10 Blue Links
Example	"Flight prices to NY"



Agentic Search

Goal	Complete Tasks
Interaction	Read & Write (Act)
Output	Final Result
Example	"Book the 5pm flight"

Key Enabler: LLMs (Reasoning) + Tools (APIs)

ReAct Pattern: Reason + Act



The Paper

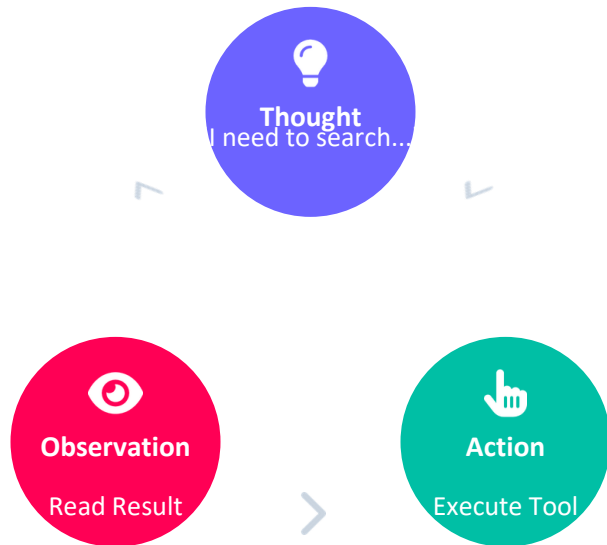
"ReAct: Synergizing Reasoning and Acting in Language Models" (Yao et al., ICLR 2023).

The Problem

LLMs can reason Chain of Thought (CoT) but can't update facts. Agents can act but may not reason well. ReAct combines both.

The Loop

The model generates a **Thought**, takes an **Action**, and receives an **Observation** from the environment.



Tool Use: APIs as Skills



⚠ The Limitation

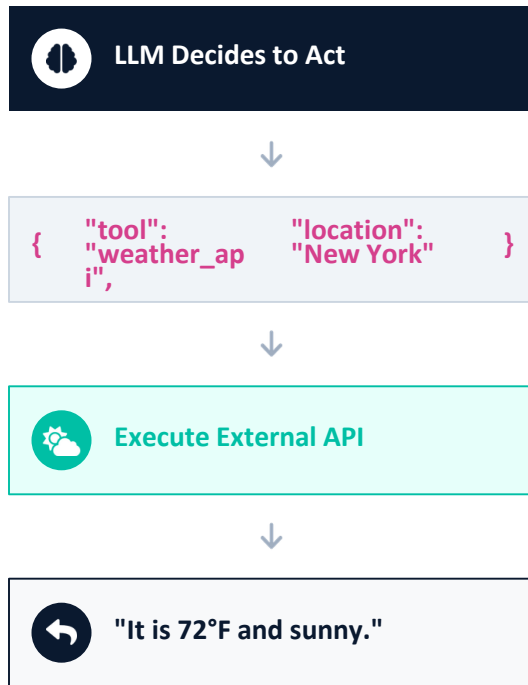
LLMs are "frozen in time" and struggle with precise math. They are text generators, not calculators.

🔧 The Solution

Give the model "hands". Define a set of tools (Calculator, Search, Calendar) it can call via structured output (JSON).

</> Function Calling

The model outputs: `call_tool("calculator", "25 * 4")` instead of guessing the answer.



Autonomous Web Agents



👁 Vision-Based Navigation

Modern agents (like GPT-4V) "see" the screen. They identify buttons and inputs visually, not just by code.

🛠 Action Space

`Click(x, y)`, `Type(text)`, `Scroll(down)`. The agent translates high-level goals into low-level UI events.

🚀 Examples

MultiOn, **Adept**, **AutoGPT**. Capable of booking flights, ordering food, and managing CRMs.

The screenshot shows a web browser window with the URL `https://fly-cheap.com/search`. The page title is "Find Flights". Below the title, there are three input fields: "From: New York" (highlighted with a red box and "ID: 1"), "To: London" (highlighted with a red box and "ID: 1"), and a "Search" button (highlighted with a blue box and "ID: 1"). Below these fields, there are two flight results listed in light blue boxes: "Flight 102 - \$450" and "Flight 405 - \$520". A mouse cursor is shown clicking on the "Search" button.

> Action: Click(ID: 14)

Neuro-Symbolic AI



Neuro-Symbolic AI



Neural (LLMs)

Pros: Fluent, creative, handles ambiguity well.

Cons: Hallucinates, bad at math/logic, opaque ("black box").

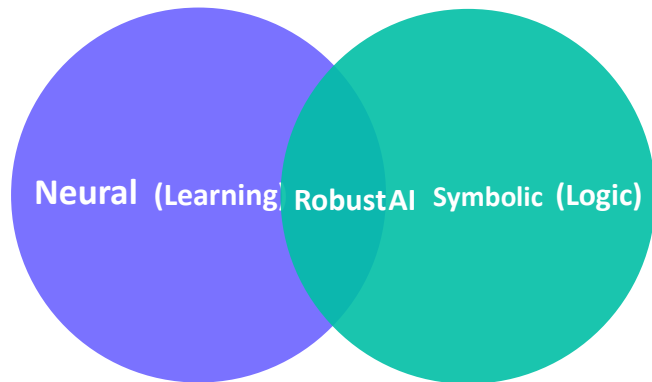
Symbolic (logic, rules, Knowledge Graphs)

Pros: Factual, logical, explainable, consistent.

Cons: Rigid, brittle, hard to scale manually.

The Synthesis

Using LLMs to query Knowledge Graphs (GraphRAG), combining **Reasoning** with **Facts**.



Knowledge Graphs (KG) Recap



The Entity

Nodes represent real-world objects (People, Places, Concepts), not just strings of text.

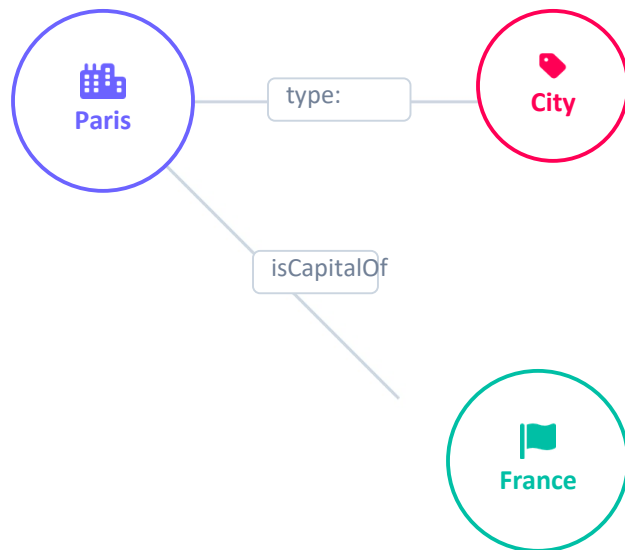
The Triple

The fundamental unit of data: **(Subject, Predicate, Object)**.

Example: (Paris, isCapitalOf, France)

The Graph

Connecting billions of triples creates a massive web of knowledge (e.g., Google Knowledge Graph, Wikidata).



GraphRAG: Structured Retrieval



The Limitation

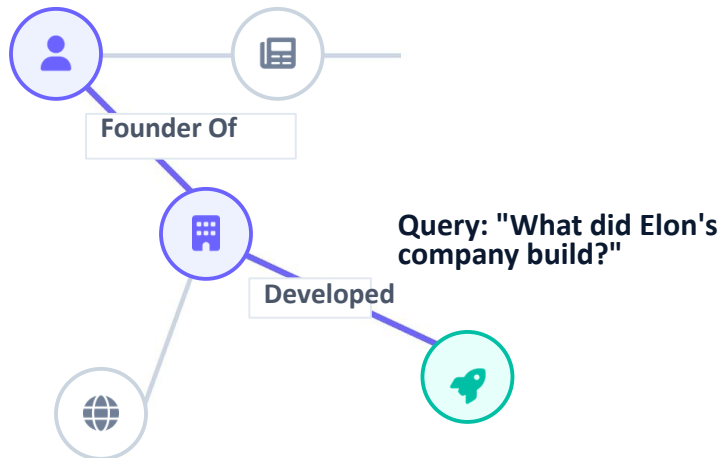
Standard RAG retrieves isolated text chunks. It struggles with multi-hop questions like "How are X and Y related?".

The Solution

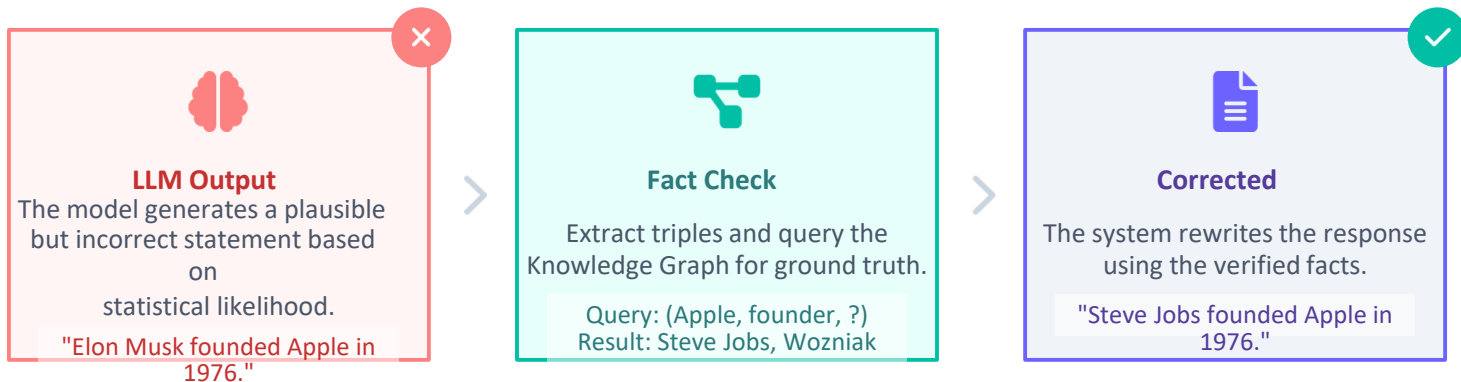
Retrieve **subgraphs** (Entities + Relationships) instead of just text.

Benefit

Provides the LLM with structured context, enabling it to "connect the dots" explicitly.



Fixing Hallucinations



"LLMs provide the fluency, Knowledge Graphs provide the facts."

Advanced Reasoning: CoT



Standard Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?



Output

The answer is 10.

✗ Incorrect (Direct Guess)

Chain-of-Thought (CoT)

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Let's think step by step.



Output

1. Roger started with 5 balls.
2. $2 \text{ cans} * 3 \text{ balls/can} = 6 \text{ balls}$.
3. $5 + 6 = 11$.

The answer is 11.



Correct (Reasoning Trace)

Tree of Thoughts (ToT)



Search, Not Just Gen

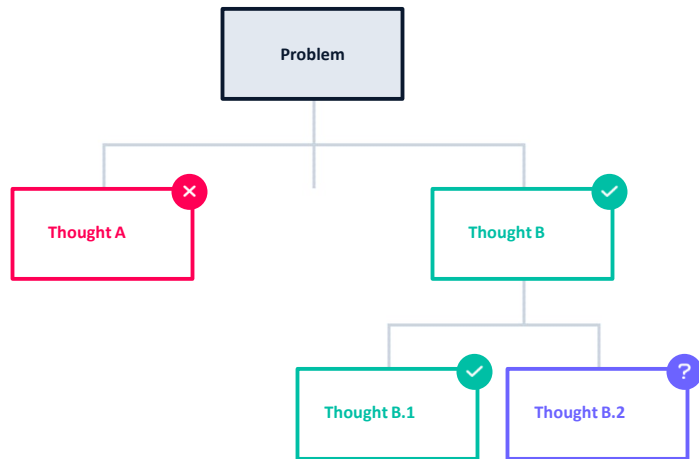
Treats reasoning as a search problem over a space of thoughts. Uses BFS or DFS to explore.

Self-Evaluation

The model generates multiple candidates and scores them: "Is this path promising?"

Backtracking

If a thought leads to a dead end, the model can discard it and return to a previous state.



Tree of Thoughts (ToT)



Real-World Uses of ToT

1. Complex Math Problems

Trying multiple solving strategies.

2. **Coding** Exploring different algorithmic solutions.

3. Planning

Travel planning, scheduling, robotics.

4. Puzzle Solving

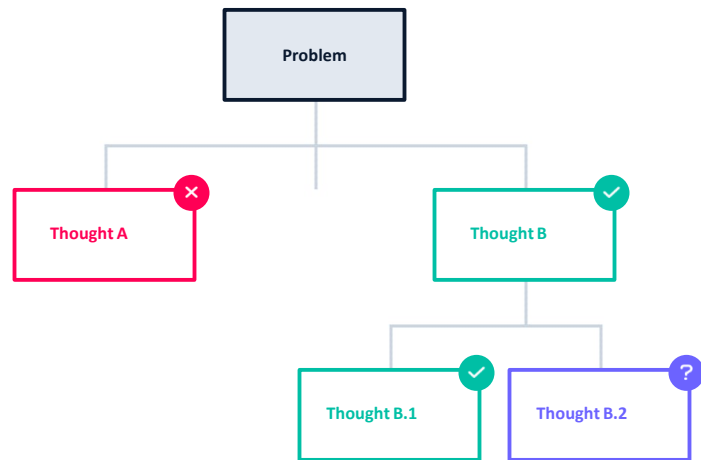
Sudoku, chess-like reasoning, logic puzzles.

5. AI Agents

Choosing between multiple actions/tools.

6. Scientific Reasoning

Testing hypotheses before selecting the best one.



Hyper-Personalization



Search History



Location



Device Type



Time of Day



Alice

Software Engineer

Query: "Python"

[Welcome to Python.org](#)

The official home of the Python Programming Language. Download the latest version...

[Python Tutorial - W3Schools](#)

Learn Python from scratch. Variables, loops, functions...



Bob

Herpetologist

Query: "Python"

[Python \(genus\) - Wikipedia](#)

Python is a genus of constricting snakes in the Pythonidae family native to the tropics...

[Ball Python Care Guide](#)

Everything you need to know about keeping a Ball Python as a pet...

Privacy-Preserving Search



The Problem

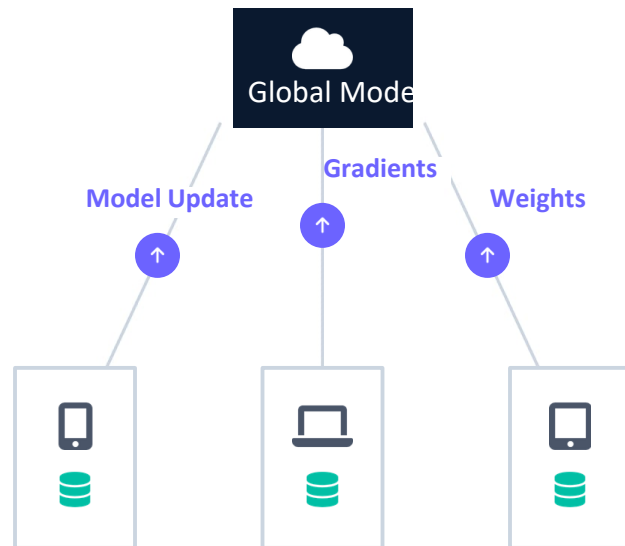
Personalization requires data, but uploading search history to the cloud risks privacy breaches.

Federated Learning

Train the model **locally** on your device. Only the learned patterns (gradients) are sent to the server.

The Guarantee

"Your raw data never leaves your phone." The global model gets smarter without seeing anyone's secrets.



Zero-UI: Ambient Computing



👁 Invisible Interface

The disappearance of the screen. Interaction happens naturally through voice, gestures, or gaze. *(No regular inputs through typing/clicking... etc)*

📍 Context-Aware

Devices know **where** you are and **what** you see. "What is this building?" becomes a valid query.

⚠ The Challenge

Zero Tolerance for Error. Without a list of 10 links to choose from, the AI must get the *one* right answer.



AI Pin

Laser Projection



Smart Glasses

Heads-Up Display



Hearables

Whisper in Ear



Rabbit R1

Action Button



"Hey, book a ride home."

Spatial Search (AR/VR)



The World is the Query

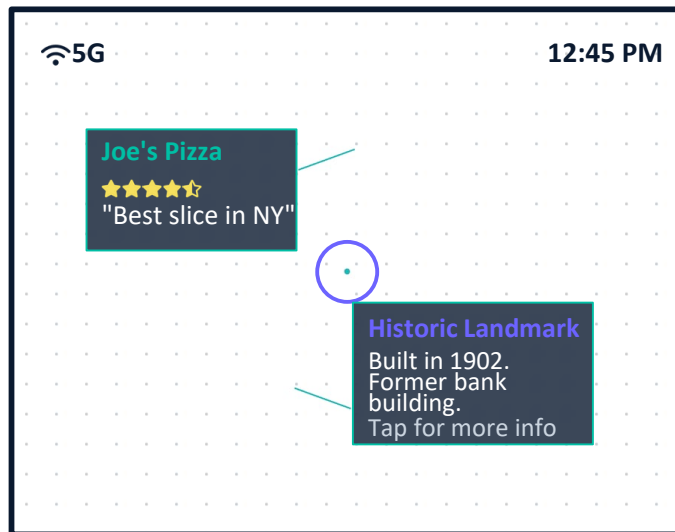
Instead of typing keywords, you look at an object. The device searches for what you see.

Contextual Overlay

Search results are "pinned" to physical locations (e.g., restaurant ratings floating above the door).

Hardware

Apple Vision Pro, Meta Ray-Ban. Moving from "Heads-Down" (Phone) to "Heads-Up" (Glasses).



Simulated AR View

Phase 1: Classic IR

Weeks 1-5

01 ▼ Text Processing

- **Tokenization:** Breaking text into words.
- **Normalization:** Case folding, Stopwords.
- **Stemming/Lemmatization:** Reducing to root forms (Porter Stemmer).
- **Zipf's Law:** Frequency distribution of terms.

03 ✓ Boolean Model

- **Exact Match:** AND, OR, NOT logic.
- **Binary:** Document is either relevant (1) or not (0).
- **Pros:** Simple, predictable.
- **Cons:** No ranking, "Feast or Famine".

02 1≡2 Inverted Index

- **Dictionary:** Terms in memory.
- **Postings Lists:** DocIDs on disk.
- **Skip Pointers:** Faster merging.
- **Compression:** Variable Byte, Gamma codes.

04 ≡ Vector Space Model

- **TF-IDF:** Term Frequency * Inverse Doc Frequency.
- **Cosine Similarity:** Measuring angle between query and doc vectors.
- **Ranking:** Sorting by relevance score.
- **SMART Notation:** Inc.ltc schemes.

Phase 2: Web Scale

Weeks 6-8



Evaluation

01

Precision, Recall, F1-Score

MAP (Mean Average Precision)

NDCG (Normalized Discounted Cumulative Gain)

User Studies & A/B Testing



Web Crawling

03

URL Frontier & Politeness

DNS Resolution & Parsing

Distributed Crawling

Duplicate Detection (Shingling)



Link Analysis

02

The Web as a Graph

PageRank (Random Surfer Model)

HITS (Hubs & Authorities)

Link Spam & TrustRank



Indexing Arch

04

MapReduce for Index Construction

Sharding (Document vs. Term)

Dynamic Indexing

Tiered Indexes

Phase 3: Neural & AI



Week 9

Neural IR & RAG

- **Embeddings:** Dense vectors (BERT) capture semantic meaning.
- **RAG:** Retrieving facts to ground LLM generation.
- **Vector DBs:** HNSW for fast similarity search.



Week 10

Recommender Systems

- **Collaborative Filtering:** Matrix Factorization (SVD).
- **Deep Learning:** Two-Tower models (YouTube).
- **Personalization:** Solving the "Long Tail" problem.



Week 11

Conversational AI

- **Dialogue Systems:** Task-oriented vs. Open-domain.
- **LLMs:** Transformers, RLHF, and Instruction Tuning.
- **Safety:** Handling hallucinations and bias.



Week 12

Future of Search

- **Multimodal:** Searching with images and video (CLIP).
- **Agents:** Autonomous systems that take action.
- **Neuro-Symbolic:** Combining Logic (KG) with Learning (LLM).

“

***"Search is not a solved problem.
It is just beginning."***

The Era of Agentic AI

Thank You

Good Luck on the Final Exam!